

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

University Examination

Date: 07/12/2017 Day: Thursday

Time: 10:00 am to 10:30 am

Total Marks: 20

Student's ID No.									
Student's Name: _____									
Student's Sign : _____									
Supervisor's Name : _____									
Supervisor's Sign: _____									

Answer Sheet for Multiple Choice Questions (MCQs), True or false and Match the following
Examination : MPT- December 2017
Semester: [3 rd]
Section : I
Course Code : PT912
Course Name: Adult Neurophysiotherapy

Important Instructions:

- There are two sections in the paper. **SECTION –I** contains 20 Questions and **SECTION – II** contains 3 Questions.
- **SECTION –I** should be answered in the question paper itself & **SECTION – II** should be answered in separate answer book.
- Marks on the right side suggest the total marks of that question.
- Maximum time allotted for **SECTION – I** is 30 minutes. Question paper for **SECTION – II** will be given once the **SECTION – I** is completed within the stipulated time and returned. However, Candidates who complete **SECTION – I** earlier than 30 minutes can collect main question paper for **SECTION –II** immediately after handing over the **SECTION-I** question paper with answers.
- **SECTION – I** should be attached with the main answer book at the end of the examination and should be submitted to the supervisor.

SECTION-I

Multiple Choice Questions: (MCQs)
(Encircle the correct answer)

(10 × 1 = 10)

1. Often the first signs of neurological disorders are deficits in basic cognitive functions and also deficits in skills that involve problem-solving, planning and engaging in goal-directed behaviour. These types of functions are known as
 - a. Directive function
 - b. Executive function
 - c. Management function
 - d. Psychological function
2. If an individual suffers a traumatic event and has a neurological disorder that means they may be unable to recall anything from the moment of the injury or to retain memories of recent events. This is known as:
 - a. Anterograde amnesia
 - b. Retrograde amnesia
 - c. Global amnesia
 - d. Dementia

(P.T.O)

3. In brain injury a severe trauma in which the brain is not just jarred but the impact also causes bruising to the brain is known as
 - a. Contusion
 - b. Concussion
 - c. Encephalitis
 - d. Cranitis
4. A 33-year-old chronically spinally injured patient becomes hypertensive and sweaty. His level of spinal cord injury is T4 and it is complete. You consider the diagnosis of autonomic hyperreflexia. Autonomic dysreflexia
 - a. Is common in a T4 lesion
 - b. Is unlikely with a T4 lesion
 - c. Should be treated by administration of beta blockers
 - d. Should be treated by opioids
5. Common complications so spinal cord injury includes all of the following except.....
 - a. Deep vein thrombosis
 - b. Heterotrophic calcification
 - c. Urinary tract infection
 - d. Diarrhea
6. The polyneuropathic pattern of sensory loss suggests presence of one of the following syndrome:
 - a. numbness & pain in distal parts of extremities
 - b. numbness & analgesia in half of the body
 - c. pain & sensory ataxia in half of the body
 - d. analgesia & sensory ataxia in proximal parts of extremities
7. In initial stage of Parkinson disease the most typical involuntary movement is the following:
 - a. chorea
 - b. atetosis
 - c. tremor
 - d. dystonia
8. A 45-year-old female retail manager has a primary complaint of numbness and tingling on the outside portion of her right hand, usually around the 4th and 5th digit. Her symptoms are aggravated after an hour of working at her desk and progressively increase as the day goes on. During the objective exam, the therapist notes that the patient has 70° of right cervical rotation ROM, minimal pain in quadrant position, and an increase in symptoms with her right arm overhead. In addition, it is noted she has an elevated 1st rib on the right compared to the left and has exquisite tenderness along her scalene musculature. What would be the appropriate initial diagnosis for this patient?
 - a. Carpal Tunnel Syndrome
 - b. Cervical Radiculopathy
 - c. Cubital Tunnel Syndrome
 - d. Thoracic Outlet Syndrome

(P.T.O)

9. A patient with a stroke affecting the right middle cerebral artery has difficulty walking, especially over uneven surfaces. Which of the following describes the MOST appropriate initial treatment to improve the patient's ability to walk over uneven surfaces?
- Place a single point cane in the patient's left hand and train him to use a step-to gait pattern.
 - Place a single point cane in the patient's right hand and train him to use a step-to gait pattern.
 - Fit the patient with a 4-wheeled walker and instruct him to use a 4-point gait pattern.
 - Fit the patient with axillary crutches and instruct him to use a 4-point gait pattern.
10. A patient who has suffered a cutting injury to the entire left half of the spinal cord at the T8 level is being examined by a physical therapist. Which of the following impairments would be MOST apparent on the ipsilateral lower extremity?
- Loss of coordination and accuracy.
 - Loss of pain and temperature sensation
 - Loss of movement and light touch sensation
 - Loss of peripheral smooth muscle control

Identify the sentence as True or False:

(5 × 1 = 5)

(Please tick mark the answer)

True False

- | | | |
|---|--------------------------|--------------------------|
| 11. Ataxia is a global term comprising inaccuracy and decomposition of movement. | ☐ | ☐ |
| 12. Entrapment at the Cubital tunnel would typically leave loss of sensation on the dorsum of the hand on the ulnar side. | ☐ | ☐ |
| 13. Difficulty in adducting and depressing the eye is indicative of Trochlear nerve involvement (CN IV) | ☐ | ☐ |
| 14. Extraocular muscle weakness or ptosis is present initially in 50% of patients and occurs during the course of illness in 90% of patients with Myasthenia gravis | ☐ | ☐ |
| 15. The gait ataxia with both sensory and cerebellar type combination is referred to as a tabetocerebellar gait | ☐ | ☐ |

Match the following

(5 × 1 = 5)

16.	Confused/Agitated: Maximal Assistance	A. Level IV
17.	Confused, Inappropriate Non-Agitated	B. Level V
18.	Confused, Appropriate Moderate Assistance	C. Level VI
19.	Automatic, Appropriate: Minimal Assistance for Daily Living Skills	D. Level VII
20.	Purposeful, Appropriate: Stand-By Assistance	E. Level VIII
		F. Level IX
		G. Level X

(16 -), (17-), (18-), (19-), (20-)

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

3rd Semester of M.P.T. Examination

December 2017

PT912 Adult Neurophysiotherapy

Date: 70/12/2017

Time: 10:30 am -1:00 pm

Maximum Marks: 50

Instructions:

1. There are two Sections in the paper. **SECTION –I** contains 20 Questions and **SECTION – II** contains 3 Questions.
2. **SECTION –I** should be answered in the question paper itself & **SECTION – II** should be answered in separate answer book.
3. Marks on the right side suggest the total marks of that question.
4. Maximum time allotted for **SECTION – II** is 2 Hr 30 min.
5. Draw the figure where necessary.

SECTION-II

Q. 1. Short answers (5 out of 7) (Answer not exceeding 50 words) (5 × 2 = 10)

- a. Risk factors of stroke.
- b. Complication of spinal cord injury.
- c. Classification of coma according to GCS.
- d. Brandt Daroff exercise for vertigo.
- a. Mechanisms of traumatic brain injury
- e. Calculate the target HR at 60% of MHR with RHR of 80bpm for a 45 year old.
- f. Name any 2 tests/ measures useful to quantify the functional ambulatory capacity in acute neurological patients.
- g. Classification of polyneuropathy.

Q. 2. Short Notes (4 out of 5) (Answer not exceeding 150 words) (4 × 5 = 20)

- b. Write a note on the multidisciplinary care for the rehabilitation of neurological disorders.
- c. Steps of clinical decision making in Neurorehabilitation.
- d. Write a note on assistive and adaptive devices used in Neurorehabilitation.
- e. Patient centered care for a patient with Myesthenia Gravis.
- f. Evidence based management of Bell's palsy.

Q. 3. Essay (2 out of 3) (2 × 10 = 20)

- a. A 25 year athlete had sustained an incomplete spinal cord injury at T10 making him wheelchair bound since 1 year. He is classified as meeting the minimum requirements of disability to participate in Paralympic sports. Describe the training to increase the strength and endurance in his upper body.

(P.T.O)

- b. A 52 year old female suffering from severe progressive multiple sclerosis complains of stiffness and increased fatigue and breathlessness during her ADL. Describe the assessment and management of fatigue to optimize her daily functions.
- c. A 17 year old girl presented with imbalance in walking since 2 years with tendency to reel and fall in an unpredictable direction, along with in-coordination of both hands since same duration, causing difficulty in writing, eating, and operating appliances. The symptoms were progressive in nature with no other remarkable findings except a positive family history. Discuss the possible diagnoses. Write the management strategy to reduce the falls.